

83% of Our Kids Are “Not College Ready” in Math. Really?



Let's just end the charade that this is a problem for students.

If **83% of first-year college students are placed into algebra or below**, that doesn't mean 83% of kids have forgotten how to do math. It means our education system is creating this outcome.

Currently, **between 40-60% of incoming college students are placed into remedial math**. But what's even more disturbing is that **less than half of those students will complete a college-level gateway math course** within two years. Remedial math courses push back graduation dates, add to students' debt, and significantly reduce the chances of obtaining a degree.

So, if your child wants to be a nurse, engineer, or scientist, math placement can quietly close that door before college even starts.

And we continue to be shocked.

The System Is Working Exactly As Designed

In *Learning to Improve*, Bryk et al. (2015) remind us that “every system is perfectly designed to get the results it gets.” If 83% of students are landing in remedial math, then that's exactly what our current math system is built to produce.

Currently, many of our classrooms are still teaching:

- Closed problems with one right answer
- Speed is more important than reasoning
- Memorization is preferred to understanding

- Three years of math, while many other states require four

Students are getting good grades. Parents are satisfied. Then they take a placement exam such as Accuplacer or SAT, and a different picture is painted.

That is not a failing student. That is a grade level and instruction level mismatch.

It's Not About Credentials — It's About Tasks

Kenney, Hiebert, and Morris (2014) stated that the improvement of a student's mathematical skill is significantly based on the quality of the task he or she is involved in. This means that if a student is only involved in repetitive procedures throughout his or her school years, he or she will never be able to develop a conceptual understanding of what he or she is learning.

Reasoning, modeling, and flexible thinking are needed in college mathematics. However, most of the time, high school rewards speed and accuracy in solving problems.

We are not surprised anymore when a student fails to answer a complex reasoning task because he or she only practiced solving problems through a worksheet.

Other Professions Figured This Out. Why Haven't We?

In the TED Talk, "How Do We Heal Medicine? By Atul Gawande," the speaker describes how improvement occurs when systems work together and improve their processes, not when people are pointed out as problems. Medicine did not get better because doctors began working harder. It got better because systems changed.

The same change is needed in education.

When 7 out of 10 students are considered "not ready," it is time for a system change, not a placement test.

What Would Actually Move 83% to Below 20%?

If we are serious about reducing remediation, we need:

Four years of significant, rigorous math in high school

- **Curriculum designed around rich, open-ended math experiences**
- Professional development that improves instruction
- Alignment of high school grading and college placement standards
- Public reporting of remediation rates by district

This is not about lowering standards.

It's about finally teaching them.

Take Action

New Jersey parents, ask your district:

- What percentage of our graduates are being placed in remedial math?
- What percentage of our graduates are completing college-level math courses within two years?
- Are our kids being challenged with rich problems or merely filling out worksheets?
- Are we connecting our high school math to our college math?

When 83% of our kids are told they are “Not ready,” maybe we should stop blaming our kids and start redesigning the system.

When the majority of our kids are failing the test, maybe we should remediate the test or the preparation.

References

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Kenney, P. A., Hiebert, J., & Morris, A. K. (2014). *Improving mathematics instruction*. National Council of Teachers of Mathematics.

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